Lab 7: Fuzzing for Input Validation Bugs

Software Engineering | SRN: PES1UG24CS634

Module: Traffic Fine Calculator (traffic_fine.py) | Tool: Hypothesis + pytest

Summary

Function	Bug found by fuzzing	Fix
parse_ticket_id	Crashes on '', wrong part count, None	Type check, exact 3 parts, validate digits; return None
total_fines	Crashes on '', '1,,3', None	Validate every item as digits; return None
speed_excess	ZeroDivisionError on '0/0', '72/0'; crashes on '/', '', None	Reject limit 0 and bad format; return None

Step 2: Run the buggy application

Command: python buggy/traffic_fine.py (input: TF-2024)

The original script crashes with a ValueError on a malformed ticket ID.

```
PS C:\Users\SUNIL\SE-PES1UG24CS634\LAB7> pip install hypothesis pytest
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\SUNIL\SE-PES1UG24CS634\LAB7> python buggy/traffic_fine.py
==== Running parse_ticket_id ====
Enter a ticket ID (e.g. TF-2024-0456): TF-2024
Traceback (most recent call last):
  File "C:\Users\SUNIL\SE-PES1UG24CS634\LAB7\buggy\traffic_fine.py", line 33, in <module>
    main()
    ~~~~^^
  File "C:\Users\SUNIL\SE-PES1UG24CS634\LAB7\buggy\traffic_fine.py", line 19, in main
    result = parse_ticket_id(user_input)
  File "C:\Users\SUNIL\SE-PES1UG24CS634\LAB7\buggy\traffic_fine.py", line 2, in parse_ticket_id
    prefix, year, serial = ticket_id.split('-')
    ^^^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 3, got 2)
PS C:\Users\SUNIL\SE-PES1UG24CS634\LAB7>
```

Step 4: Fuzz tests failing on the buggy code

Command: python -m pytest test_traffic_fine.py

Hypothesis found crashing inputs in all three functions. Result: **4 failed, 3 passed**.

```
PS C:\Users\SUNIL\SE-PES1UG24CS634\LAB7> cd C:\Users\SUNIL\SE-PES1UG24CS634\LAB7
PS C:\Users\SUNIL\SE-PES1UG24CS634\LAB7> mkdir ..\temp_buggy


    Directory: C:\Users\SUNIL\SE-PES1UG24CS634


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        13-11-2026     16:47                temp_buggy

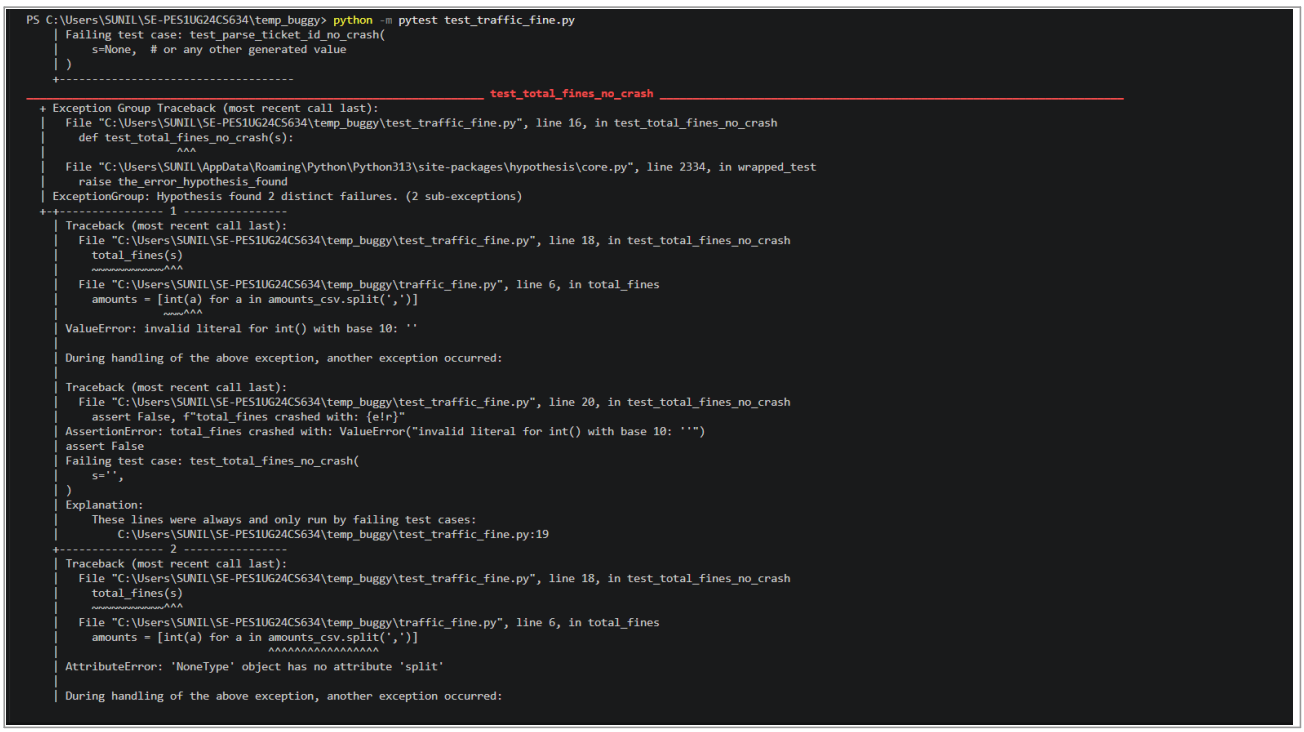

PS C:\Users\SUNIL\SE-PES1UG24CS634\LAB7> copy buggy\traffic_fine.py ..\temp_buggy\
PS C:\Users\SUNIL\SE-PES1UG24CS634\LAB7> copy test_traffic_fine.py ..\temp_buggy\
PS C:\Users\SUNIL\SE-PES1UG24CS634\LAB7> cd ..\temp_buggy
PS C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy> python -m pytest test_traffic_fine.py
============================= test session starts ==============================
platform win32 -- Python 3.13.0, pytest-9.1.1, pluggy-1.7.0
rootdir: C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy
plugins: anyio-4.14.2, hypothesis-6.170.15, cov-7.1.0
collected 7 items

test_traffic_fine.py FFF...F                                             [100%]
```

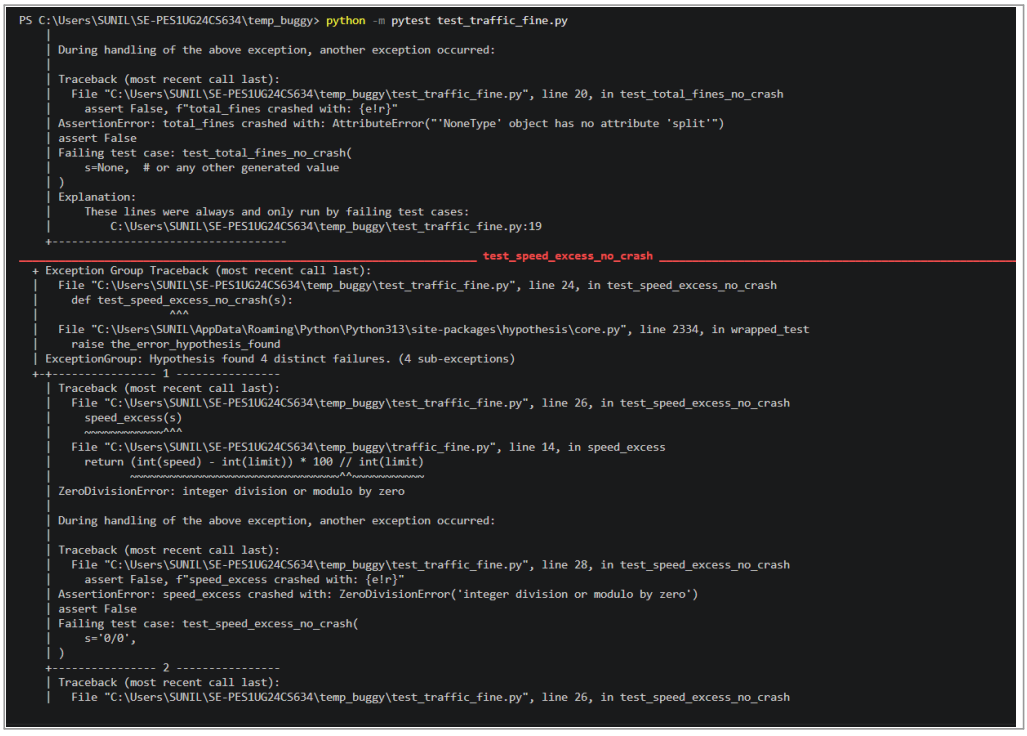
Step 4, screenshot 1 of 6

```
PS C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy> python -m pytest test_traffic_fine.py
_____ test_parse_ticket_id_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy\test_traffic_fine.py", line 8, in test_parse_ticket_id_no_crash
  |     def test_parse_ticket_id_no_crash(s):
  |                ^^^
  |   File "C:\Users\SUNIL\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy\test_traffic_fine.py", line 10, in test_parse_ticket_id_no_crash
    |     parse_ticket_id(s)
    |     ~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy\traffic_fine.py", line 2, in parse_ticket_id
    |     prefix, year, serial = ticket_id.split('-')
    |     ^^^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 3, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy\test_traffic_fine.py", line 12, in test_parse_ticket_id_no_crash
    |     assert False, f"parse_ticket_id crashed with: {e!r}"
    | AssertionError: parse_ticket_id crashed with: ValueError('not enough values to unpack (expected 3, got 1)')
    | assert False
    | Failing test case: test_parse_ticket_id_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy\test_traffic_fine.py", line 10, in test_parse_ticket_id_no_crash
    |     parse_ticket_id(s)
    |     ~~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy\traffic_fine.py", line 2, in parse_ticket_id
    |     prefix, year, serial = ticket_id.split('-')
    |                            ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\SUNIL\SE-PES1UG24CS634\temp_buggy\test_traffic_fine.py", line 12, in test_parse_ticket_id_no_crash
    |     assert False, f"parse_ticket_id crashed with: {e!r}"
    | AssertionError: parse_ticket_id crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_parse_ticket_id_no_crash(
```

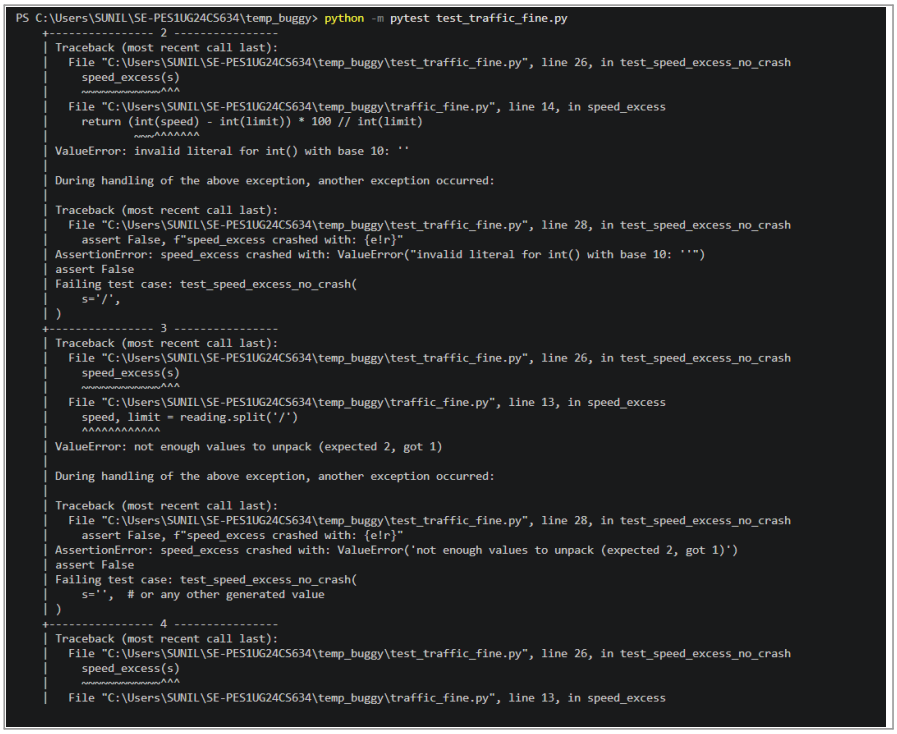
Step 4, screenshot 2 of 6



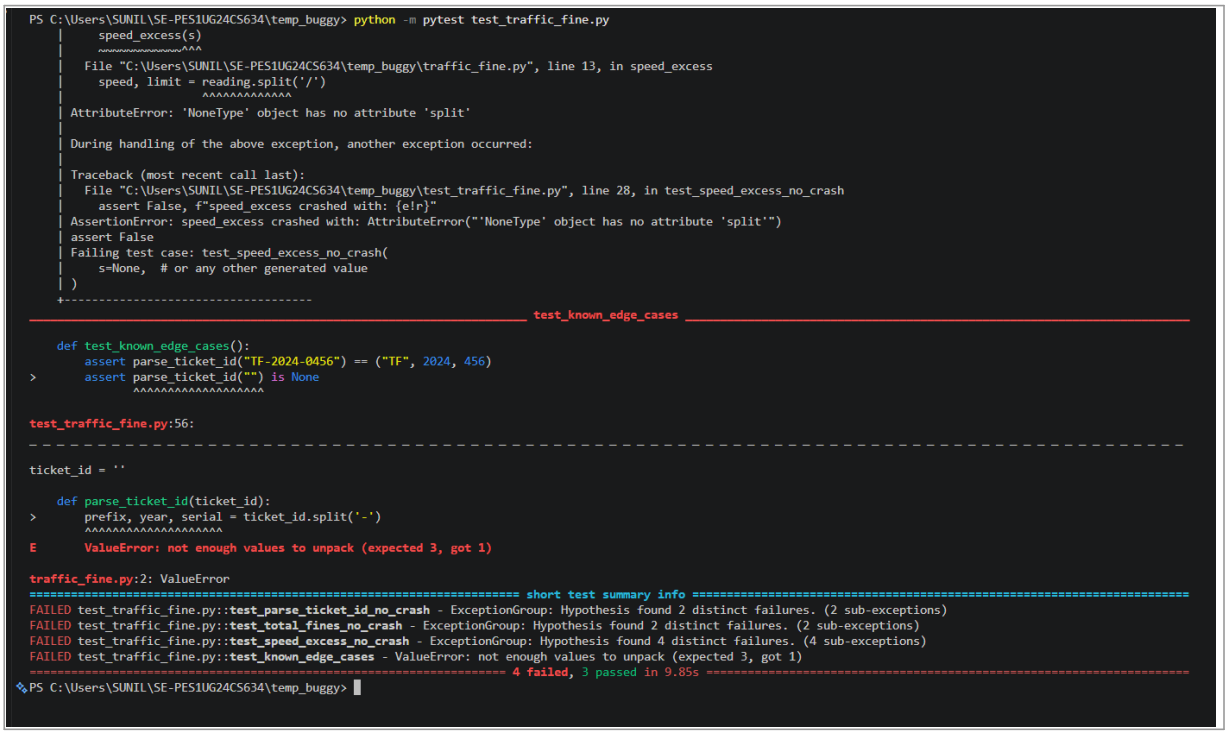
Step 4, screenshot 3 of 6



Step 4, screenshot 4 of 6



Step 4, screenshot 5 of 6



Step 4, screenshot 6 of 6

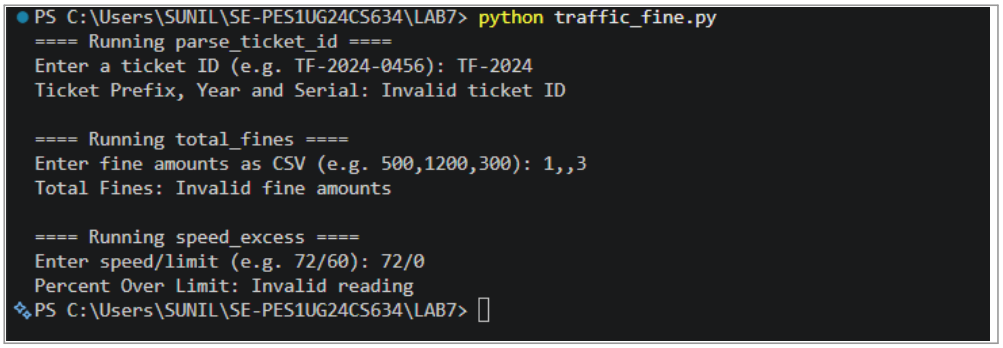
Step 5: Run the fixed application

Command: python traffic_fine.py

(a) Valid input: TF-2024-0456, 500,1200,300, 72/60



(b) Invalid input: TF-2024, 1,,3, 72/0. The fixed script reports invalid input instead of crashing.



Step 6: Re-run the fuzz tests on the fixed code

Command: python -m pytest test_traffic_fine.py

Result: **7 passed**.

